THE UNITED REPUBLIC OF TANZANIA

PRIME MINISTER'S OFFICE, REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

MOROGORO REGIONAL FORM TWO TERMINAL EXAMINATION

031 **PHYSICS**

TIME: 2:30Hrs **YEAR: 2026**

Instructions

1. This paper consists of sections A, B and C with a total of **TEN (10)** questions.
2. Answer **ALL** questions in ALL sections.
3. Mathematical tables may be used.
4. Any communication device and any unauthorized material are **not** allowed in the examination room.
5. Write your **Examination Number** on every page of your answer booklet(s).
6. Where necessary the following constants may be used:
   (i) Acceleration due to gravity, g = 10 m/s^2
   (ii) Density of water = 1.0 g/cm^3 or 1000Kg/m^3
   (iii) Pi, π = 3.14.

FOR EXAMINER'S USE ONLY		
QUESTION NUMBER	**SCORE**	**EXAMINER'S INITIALS**
1		
2		
3		
4		
5		
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7		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

SECTION A (16 Marks)

1. Choose the most correct answer from the given alternatives and write its letter in the answer booklet provided; -

i. The pressure of water in a dam is greater at the bottom, which of the following is a reason as to why a dam has thick walls at the bottom?
   A. To counter- balance the pressure
   B. To accommodate more water
   C. To reduce energy generation
   D. To increase energy generation
   E. To maintain the shape for commercial purpose.

ii. A total eclipse of the Sun is due to
   A. the Moon coming between the Earth and the Sun
   B. the Earth coming between the Moon and the Sun
   C. the Moon reflecting light away from the Earth
   D. the Sun coming between the Earth and the Moon
   E. the Earth reflecting light away from the Moon.

iii. Which of the following statements is true when the resistance, *R*, of a wire is measured using an ammeter, voltmeter and rheostat?
   A. the ammeter is in parallel with R
   B. the voltmeter is in series with R
   C. a graph of V against I has a gradient equal to R
   D. a graph of I against V has a gradient equal to R
   E. a graph of R against I will be a straight line.

iv. The force that opposes the gravitational force of a load suspended by a string is called
   A. Torsion force
   B. Extension
   C. Tension
   D. Applied force
   E. Upward force

v. Glue binds pieces of paper because of_______
   A. Adhesive force
   B. Cohesive force
   C. Capillarity
   D. Surface tension
   E. Magnetic forces

vi. An instrument which is used to observe objects around obstacles is called:
   A. Plane glass
   B. Telescope
   C. Microscope
   D. Binocular
   E. Periscope

vii. Which of the following represent a rate quantity?
   A. Ampere and volt
   B. Newton and joule
   C. Volt and watt
   D. Ampere and watt
   E. Metre/second and speed

viii. A conductor becomes an electromagnetic when

A. it is wrapped with a coil of wire
B. a soft iron core is used
C. its resistance is increased
D. an electric current flows through it
E. It is passed through an electric field

ix. A material which allows some light to pass through it but one cannot see thro ugh it is said to be
A. Transparent
B. Translucent
C. Luminous
D. Opaque
E. Colourless.

x. Which of the following scientific statements need to be proved through scientific research?
A. Hypothesis
B. Principle
C. Conclusion
D. Proposal
E. Measurement.

2. Match the descriptions of the micrometer screw gauge in List A with responses in List B by writing the letter of the correct response beside the item number **5 marks**

List A	List B
i. A movable rod of a micrometer screw gauge touching the object ii. The distance between the successive threads iii. Contained in a thimble used to tighten the body iv. Its accuracy has three decimal places in centimeter. v. Contains linear scale	A. Pitch B. Anvil C. Spindle D. Circular scale E. A lock button F. Ratchet G. Thimble H. Sleeve I. Diameter of a wire

SECTION B (70 Marks)

3. (a) (i) Identify three conditions to be satisfied before a body can float. **(03 marks)**

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(ii) Outline any three factors affecting the upthrust of the fluid. **(03 marks)**

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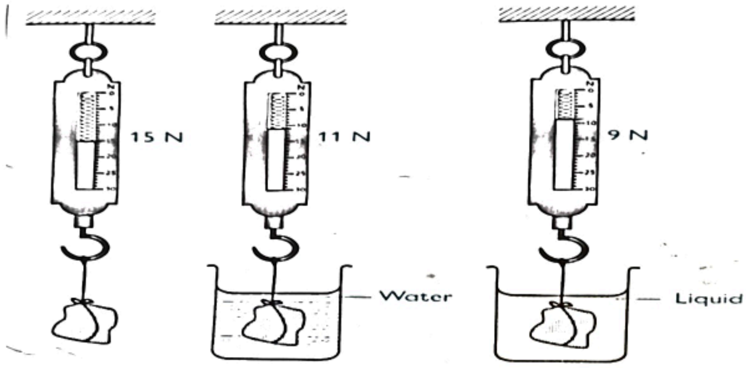
(b) Use the figure below; determine the relative density of the liquid.

(04 marks)



4. A car accelerates from a speed of 80m/s to a speed of 120m/s in 1 minute. It maintains the maximum speed reached for 20 seconds. It then decelerates uniformly to a stop after 40 seconds. It remains stationary for 20 seconds. Draw the velocity time graph for the journey. Hence from the graph calculate;

   (04 marks)

   (i) The deceleration of the car **(03 marks)**

   (ii) The total distance travelled by the car **(03 marks)**

5. (a) (i) A student was asked by his grandmother about how static electricity can be converted to current electricity. Explain how you would help the student answer correctly.

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(3 marks)

(ii) Write down the two factors that can determine the magnitude of the electric forces between the charged bodies. **(2 marks)**

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(b) Three capacitors of 2μF, 3μF and 5μF arranged in series are connected across 10V power supply. Determine the total quantity of charge stored in each capacitor. **(05 marks)**

6. (a) i. Draw the ray diagrams to show the position of the image formed when an object is placed beyond the Centre of curvature of a concave mirror. **(03 Marks)**

ii. Mention two conditions for the total internal reflection of light to occur **(2 marks)**

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(b) The focal length of a converging lens is 12cm. How far should the lens be from the image which is four times the size of the object on a screen? What is the nature of the image formed? **(05 Marks)**

7. (a) (i) Differentiate between osmosis and diffusion. **(02 marks)**

(ii) Identify two factors affecting the capillarity of a capillary tube. **(02 marks)**

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(b) A rubber band has an original length of 20 cm and is stretched to a length of 25 cm. What is the percentage change in length of the rubber band? **(06 marks)**

8. (a) Tourist went on a tour and he observed that it takes more time to cook food at the top of Mount Kilimanjaro than at its lowest level, but he did not know the reason behind. Briefly explain why it takes longer at the top of mount Kilimanjaro than at the base **(02 marks)**

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(b) A pendulum bob of mass 100g suspended with an 80cm string is raised to vertical height of 20cm from the horizontal and then released. Find;

(i) the maximum potential energy of the bob **(03 marks)**

(ii) the maximum speed of the bob **(02 marks)**

(iii) The periodic time of a bob. **03 marks)**

9. (a) What is meant by
   (i) The angle of declination? **(02Marks)**

   (ii) A magnetic pole? **(02 Marks)**

   (iii) 0utline four applications of magnets in real life **(02 marks)**

   (b) How can you test the polarity of a magnet? Explain briefly, why attraction is not a sure test for polarity. **(04 Marks)**

SECTION C (15 Marks)

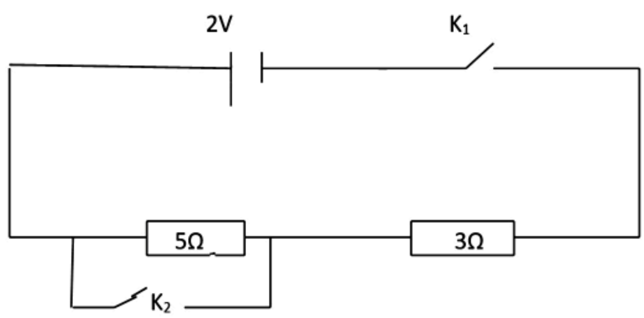
10. (a) Draw the symbols of the following electric components

i. Key **(01 mark)**

ii. Variable resistor **(01 mark)**

iii. Capacitor **(01 mark)**

iv. Fuse **(01 mark)**



b) Use the circuit above to calculate the current when

(i) Switches K_1 and k_2 are closed **(02 mark)**

(ii) Switch K_1 is closed while k_2 is open **(02 mark)**

(iv) Switch K_1 is open while switch K_2 is closed **(03 mark)**

c) state the following laws applied in electricity

i) Joule's law **(02 mark)**

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ii) ohm's law **(02 mark)**

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